Hedonic Rent Regressions: Long-Run Filing Rate

Philly Evictions Project

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1 Overview

This document summarizes the hedonic rent regressions from `r/price-regs.R`. The script estimates the relationship between eviction filing intensity (raw long-run pre-COVID filing rate) and rent levels, controlling for building and neighborhood characteristics.

Unit of analysis: Building (PID) × year panel, 2014–2019.

Key data inputs:

- `analytic_sample.csv` — PID × year panel with rents, filing rates, building characteristics, and tenant demographics

Sample restrictions:

- Years 2014–2019 (pre-COVID hedonic)
- Non-missing long-run pre-COVID filing rate
- Filing rate ≤ 0.75 (excludes extreme outliers)

2 Eviction Intensity Measures

Two representations of eviction filing intensity are used:

- **Binned:** Long-run pre-COVID filing rate cut at 0, 5%, 10%, 20%, with reference category (5–10%]
- **Continuous:** Raw long-run filing rate (pre-COVID), entering linearly

3 Empirical Strategy

3.1 Fixed Effects and Controls

All specifications include a rich set of fixed effects:

- **Year** (absorbs rent trends)
- **GEOID** (census tract — absorbs neighborhood-level rent premia)
- **Year built decade** (construction vintage)
- **Rent source** (Altos vs other)
- **Unit count bin** (building size category)
- **Building type** (structural type)
- **Quality grade** (A/B/C/D, standardized)
- **Number of stories bin**

Additional controls: log(total area), log(market value).

Regressions are weighted by total units and clustered by PID.

3.2 Model 1: Binned Filing Rate (Baseline)

$$\log(\text{rent}_{it}) = \sum_b \beta_b \cdot \mathbb{1}[\text{filing rate bin}_i = b] + \gamma_1 \cdot \text{ViolFlag}_i + \gamma_2 \cdot \text{ComplaintFlag}_i + \delta' X_i + \alpha_{\text{tract}} + \phi_t + \varepsilon_{it}$$

where ViolFlag_i indicates any unsafe/dangerous/hazardous violation and ComplaintFlag_i indicates any heat/fire/drainage/plumbing complaint. Reference bin: (5–10%].

3.3 Model 2: Continuous Filing Rate

$$\log(\text{rent}_{it}) = \beta \cdot \overline{\text{filing rate}}_i + \delta' X_i + \alpha_{\text{tract}} + \phi_t + \varepsilon_{it}$$

3.4 Models 3–4: Tenant Composition Controls

Models 1 and 2 augmented with additive tenant composition controls:

- `infousa_pct_black_imp` (share Black, mean-imputed)
- `infousa_pct_female_imp` (share female HoH, mean-imputed)
- `infousa_pct_black_female_imp` (share Black female HoH)

- `infousa_share_persons_demog_ok_imp` (demographic coverage)
- `tenant_comp_missing` (indicator for missing composition)

3.5 Model 5: Complaint × High-Eviction Interaction

$$\log(\text{rent}_{it}) = \beta_1 \cdot \text{ComplaintFlag}_i + \beta_2 \cdot \text{HighEvict}_i + \beta_3 \cdot \text{ComplaintFlag}_i \times \text{HighEvict}_i + \delta' X_i + \alpha_{\text{tract}} + \phi_t + \varepsilon_{it}$$

where $\text{HighEvict}_i = \mathbb{1}[\text{filing rate}_i > 0.20]$. Uses a sparser FE set (year + GEOID + unit count bin).

4 Results

4.1 Baseline: Binned and Continuous Specifications

Dependent Variable:	log_med_rent	
Model:	(1)	(2)
Variables		
Filing rate = 0	0.0953***	
	(0.0118)	
Filing rate (0, 5%]	0.0477***	
	(0.0130)	
Filing rate (10, 20%]	-0.0295***	
	(0.0110)	
Filing rate 20%+	-0.0377***	
	(0.0097)	
Fixed-effects		
year	Yes	Yes
GEOID	Yes	Yes
year_blt_decade	Yes	Yes
source	Yes	Yes
num_units_bin	Yes	Yes
building_type	Yes	Yes
quality_grade_standard	Yes	Yes
num_stories_bin	Yes	Yes
Fit statistics		
Observations	116,641	116,641
R^2	0.67288	0.66984

Clustered (PID) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

4.2 Full Table: With Tenant Composition Controls

Dependent Variable:	log_med_rent			
Model:	(1)	(2)	(3)	(4)
Variables				
Filing rate = 0	0.0953***		0.0911***	
	(0.0118)		(0.0117)	
Filing rate (0, 5%]	0.0477***		0.0463***	
	(0.0130)		(0.0129)	
Filing rate (10, 20%]	-0.0295***		-0.0261**	
	(0.0110)		(0.0110)	
Filing rate 20%+	-0.0377***		-0.0343***	
	(0.0097)		(0.0096)	
any_unsafe_dangerous_violationTRUE	-0.0076		-0.0060	
	(0.0156)		(0.0155)	
any_heat_fire_drainage_plumbing_complaintTRUE	-0.0263*		-0.0244	
	(0.0150)		(0.0149)	
log(total_area)	0.0079	0.0079	0.0080	0.0080
	(0.0075)	(0.0077)	(0.0075)	(0.0076)
log(market_value)	0.0533***	0.0523***	0.0528***	0.0519***
	(0.0091)	(0.0090)	(0.0091)	(0.0090)
Filing rate (pre-2019)		-0.3270***		-0.3024***
		(0.0284)		(0.0280)
infousa_pct_black_imp			-0.1143***	-0.1159***
			(0.0156)	(0.0157)
infousa_pct_female_imp			-0.0592***	-0.0610***
			(0.0101)	(0.0103)
infousa_pct_black_female_imp			0.0574***	0.0585***
			(0.0135)	(0.0136)
infousa_share_persons_demog_ok_imp			0.0327**	0.0398***
			(0.0152)	(0.0154)
tenant_comp_missing			0.0382***	0.0454***
			(0.0091)	(0.0091)
Fixed-effects				
year	Yes	Yes	Yes	Yes
GEOID	Yes	Yes	Yes	Yes
year_blt_decade	Yes	Yes	Yes	Yes
source	Yes	Yes	Yes	Yes
num_units_bin	Yes	Yes	Yes	Yes
building_type	Yes	Yes	Yes	Yes
quality_grade_standard	Yes	Yes	Yes	Yes
num_stories_bin	Yes	Yes	Yes	Yes
Fit statistics				
Observations	116,641	116,641	116,641	116,641
R^2	0.67288	0.66984	0.67480	0.67209

Clustered (PID) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

4.3 Coefficient Comparison: Eviction Intensity Across Specifications

Table 1: Eviction intensity coefficients: baseline vs tenant-composition controlled

Model	Term	Estimate	SE	p
baseline_bin	0	0.0953	0.0118	0.000
baseline_bin	(0-5%]	0.0477	0.0130	0.000
baseline_bin	(10-20%]	-0.0295	0.0110	0.008
baseline_bin	20%+	-0.0377	0.0097	0.000
tenant_bin	0	0.0911	0.0117	0.000
tenant_bin	(0-5%]	0.0463	0.0129	0.000
tenant_bin	(10-20%]	-0.0261	0.0110	0.018
tenant_bin	20%+	-0.0343	0.0096	0.000
baseline_cont	filing_rate_longrun_pre2019	-0.3270	0.0284	0.000
tenant_cont	filing_rate_longrun_pre2019	-0.3024	0.0280	0.000

4.4 Coefficient Plot: Binned Model

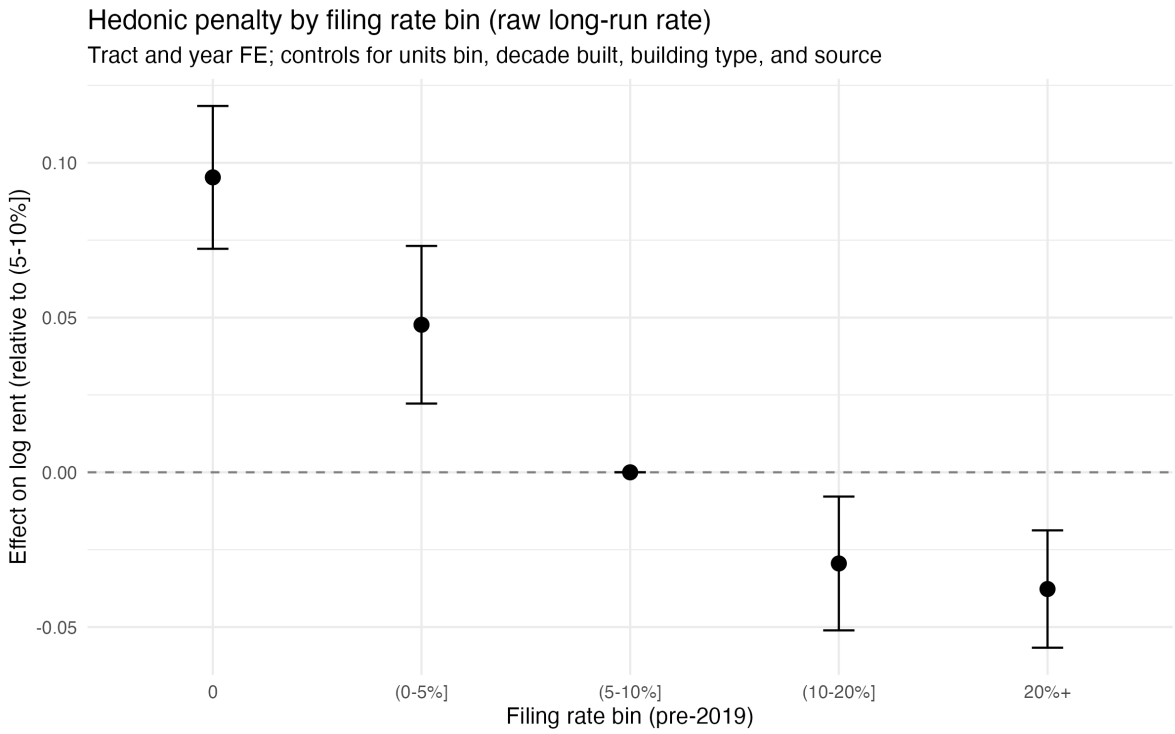

Figure 1: Hedonic rent penalty by filing intensity bin (baseline specification). Reference category: 5-10 percent long-run filing rate bin. Tract and year FE with building controls.

5 Exploratory Diagnostics

These plots residualize both rent and eviction intensity on the same controls and fixed effects used in the hedonic regressions, then examine the remaining relationship.

5.1 Residual Variation in Filing Rate

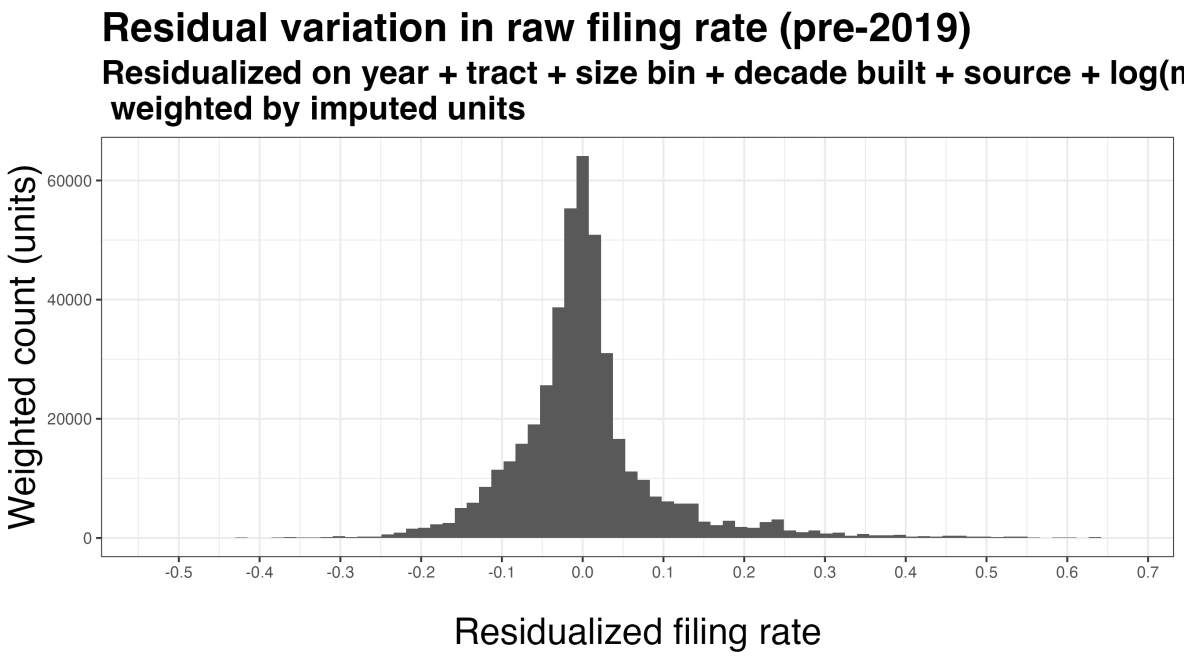

Figure 2: Distribution of residualized long-run filing rate (pre-2019). Residualized on year + tract + building controls; weighted by total units.

5.2 Residualized Rent vs Eviction Intensity: Bin Scatter

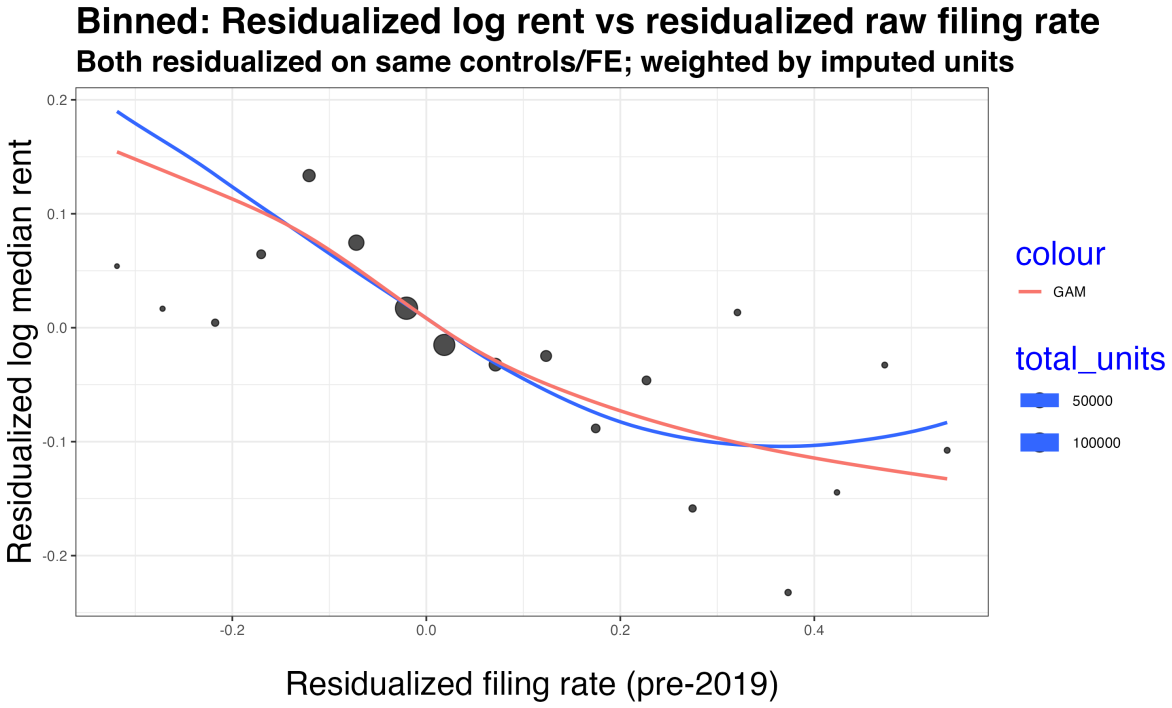

Figure 3: Binned scatter of residualized log rent vs residualized long-run filing rate. Both residualized on the same controls/FE; weighted by total units.

5.3 Spline-Implied Stigma Curve

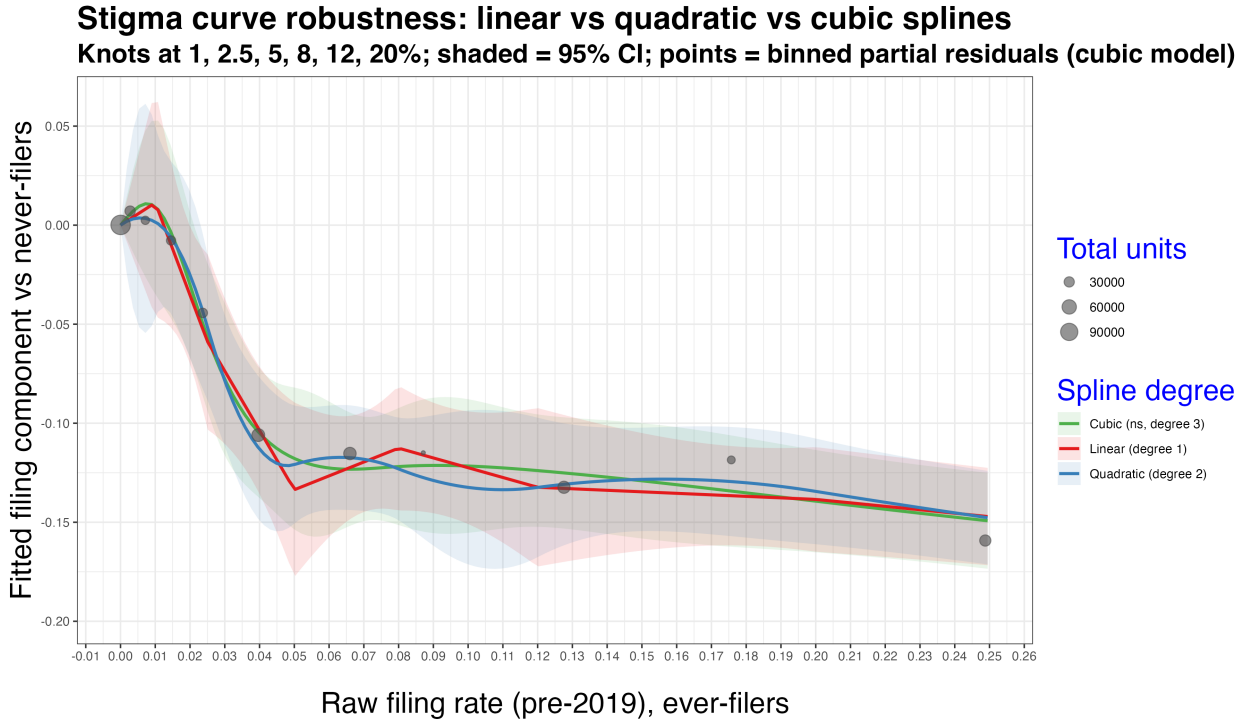

Figure 4: Spline-implied stigma curve: linear, quadratic, and cubic B-spline specifications overlaid. All three use hurdle=FALSE (spline through zero); knots at 1%, 2.5%, 5%, 8%, 12%, 20%. The similarity across degrees indicates robustness of the shape to spline flexibility.

6 Interpretation

The hedonic regressions estimate the **rent penalty** (or premium) associated with eviction filing intensity, conditional on neighborhood (tract FE) and building observables. Key quantities:

- **Binned model (Model 1):** The coefficient on the 20%+ bin relative to (5–10%] gives the rent differential for the highest-evicting buildings vs moderate-evicting buildings *within the same tract and building type*.
- **Continuous model (Model 2):** A one-unit increase in the long-run filing rate (e.g., from 0 to 1, which spans the entire range) is associated with a β-log-point change in rent. For typical variation (0 to 0.20), multiply by 0.20.
- **Tenant composition stability:** If adding tenant composition controls (Models 3–4) substantially attenuates the eviction intensity coefficients, it suggests the rent–eviction relationship partly reflects sorting by demographics. If coefficients are stable, the relationship is robust to observable tenant composition.
- **Complaint $\times$ high-eviction interaction (Model 5):** Tests whether buildings with both habitability complaints and high eviction rates have differentially lower rents, consistent with a "stigma" or quality channel.